Learning to Add, Remove and Swap: Reversible Reinforcement Learning for Excitation–Emission Pair Selection

[Author names]
[Affiliations]

***Abstract*—Selecting a few excitation–emission pairs from a multi-excitation fluorescence image is a fixed-budget combinatorial problem in which the value of a pair depends on the pairs already chosen. Constructive reinforcement-learning (RL) policies can only add, so early mistakes are permanent. We formulate pair selection as a reversible Markov decision process with add, remove and stop actions, paired credit for each remove–add swap, tabu masking, and a permutation-invariant candidate scorer built on label-free residual-variance features, so that one policy applies to datasets with different candidate sets. On four annotated datasets (two lichen, two collagen) with nested, specimen-grouped evaluation we separate three regimes. On a single labelled dataset RL is a weak black-box optimiser: simulated annealing and a genetic algorithm reach ~~the exact optimum~~ (near-)exact optima at $n = 3$ and higher objective values at a matched budget. Deployed *without labels* on an unseen sample type, the reversible policy beats a matched add-only policy by $+0.025$ balanced accuracy (pooled over targets, sources and budgets; 95% CI $[0.022, 0.028]$; significant for all four variants), the gain is largest across sample families, and the reversible policy is not harder to train. A label-free greedy ~~baseline nevertheless remains hard to beat~~ column-subset baseline (CSS) is nevertheless strong at small budgets, ~~and~~ though below random selection at $n = 13$ on three of four targets; using the swap policy to refine ~~that greedy~~ the CSS set helps on ~~one target but usually~~ Lichens 1, lowers accuracy on Lichens 2 and leaves it unchanged on both collagen targets. With 25 labelled evaluations, transferred policies match simulated annealing, which overtakes them at larger budgets.**

***Index Terms*—band selection, reinforcement learning, fluorescence, hyperspectral imaging, combinatorial optimisation**

I. Introduction

Multi-excitation fluorescence hyperspectral imaging records, for every pixel, an excitation–emission matrix (EEM): the sample is illuminated at several excitation wavelengths λ^{ex} and a full emission spectrum is recorded at each [1]. One measurement channel is therefore a *pair* $c = (\lambda^{\mathrm{ex}}, \lambda^{\mathrm{em}})$, and an acquisition yields ~~150–210~~ 149–210 admissible pairs after ~~Rayleigh-scatter rejection~~the bands nearest to the excitation wavelength and to twice the excitation wavelength (first- and second-order scattered excitation light) are removed. Measuring all of them is slow, photobleaches the sample and needs a ~~tunable source~~ switchable multi-LED source with a tunable emission filter; a handful of well-chosen pairs would allow a cheap instrument with a few LEDs and fixed filters. Choosing n of M pairs is a combinatorial problem ($\binom{192}{13} \approx 5 \times 10^{19}$), and the value of a pair depends on the pairs already chosen.

Reinforcement learning (RL) is attractive for such sequential set construction [2]–[4], and has been applied to hyperspectral band selection [5]. Constructive policies, however, can only *add*: an early mistake is permanent. In combinatorial RL, reversible actions that also remove elements improved solution quality [6], [7]. We ask whether the same holds for spectral pair selection, and, crucially, *where* RL can add value at all. Our contributions are:

- a reversible add/remove/stop MDP for fixed-budget pair selection with paired credit assignment, tabu masking and a permutation-invariant candidate scorer that transfers across datasets with different candidate sets;
- a protocol that separates three regimes—per-instance search at a matched evaluation budget, label-free transfer to an unseen dataset, and few-shot search with a small labelled budget—with nested, specimen-grouped evaluation;
- a study on four labelled fluorescence datasets (two lichen, two collagen) against classic search (greedy, swap, beam, simulated annealing, genetic), exact small-budget optima and label-free baselines.

We find that reversibility helps ~~exactly where RL is used beyond a single instance—on new data—while~~ in label-free transfer to new data (S2); with a few labelled evaluations on the target, look-ahead over best-of-k sampling brings no significant gain when pooled over datasets and budgets (S3), and classic search remains the better optimiser of a fixed objective.

II. Problem

Let $\mathcal{C} = \{c_j\}_{j=1}^{M}$ be the admissible pairs (~~first-~~ bands near λ^{ex} and ~~second-order Rayleigh bands~~ $2\lambda^{\mathrm{ex}}$ removed). Pixels $x_i \in \mathbb{R}^M$ carry a class y_i and a specimen id g_i. For a budget n we seek

$$S^{\star} = \arg\max_{S \subset \mathcal{C},\, |S|=n} \mathrm{BA}_{\mathrm{test}}(S), \tag{1}$$

the balanced accuracy of a classifier trained on the selected pairs and tested on *specimens not used for selection*. The search can only use a surrogate computed on training specimens. We use two: **Supervised** $J_{\mathrm{sup}}(S)$: the balanced accuracy of shrinkage LDA under inner whole-specimen cross-validation. Per inner fold we precompute class sums and the pooled within-class scatter for all M pairs once; scoring any

S is then a $|S| \times |S|$ submatrix solve (≈ 0.5 ms), which makes RL training with 10^5–10^6 evaluations feasible. **Label-free** $J_{\mathrm{rec}}(S) = \mathrm{tr}(\Sigma_{:S}\Sigma_{SS}^{-1}\Sigma_{S:})/\mathrm{tr}\,\Sigma$: the fraction of variance of all M standardised pairs that is linearly predictable from S (column-subset selection [8]), computable on a new, unlabelled sample. Labels are expensive and absent for new samples, so a method meant for new data must act label-free; J_{sup} is available only on annotated training datasets.

III. Reversible selection MDP

State and actions. The state is the current set S_t, a phase $q_t \in \{\textsc{add}, \textsc{remove}\}$ and a tabu list. Episodes start from $S_0 = \emptyset$ (or from a label-free greedy set) and *construct* by ADD until $|S| = n$. In *add/remove* mode they then run up to $T = n$ refinement cycles: the agent either STOPs or removes a pair $c_r \in S$, after which only an ADD of $c_a \notin S$ is allowed, so every cycle is a learned swap $S' = (S \setminus \{c_r\}) \cup \{c_a\}$ that preserves the budget. Masks forbid duplicates, invalid pairs, re-adding a removed pair for $L = 3$ cycles and removing the pair just added; revisiting a subset costs -0.01. **Reward.** Construction steps receive $r_t = J(S_{t+1}) - J(S_t)$ with $J(\emptyset)$ at chance. A removal receives 0 and the following addition $J(S') - J(S)$, so ~~both halves of a swap share the credit~~ the removal is credited only through the return of the completed ~~replacement and the~~ swap ($\gamma = 1$, GAE $\lambda = 0.95$), not by an explicit split. Without the revisit penalty the return telescopes to ~~$J(S_{\mathrm{final}}) - J(\emptyset)$~~$J(S_{\mathrm{final}}) - J(S_0)$, with $S_0 = \emptyset$ ($J(\emptyset)$ at chance) for construction episodes and S_0 the CSS set for CSS-start episodes; each revisit lowers it by 0.01. **Candidate features.** Each pair c is described, given S, by label-free quantities standardised within the task: wavelengths and Stokes shift (set to zero in the wavelength-free configuration used for all S2 and S3 results; S1 and the dashed curves of Fig. 4a use the full set), log-intensity mean and spread, residual variance of c given S, the gain in J_{rec} (and in a whitened-PCA variant) if c is added or the loss if it is removed—all from rank-one updates of the residual covariance—maximum and mean $|\mathrm{corr}|$ with S, and excitation-sharing indicators. Optional supervised features (Fisher score, exact pair-enumeration priors) are used only when labels are available. **Policy.** A shared MLP embeds every candidate; the set context is the mean and max of the selected embeddings plus global features (phase, $|S|/n$, current J_{rec}) [9]. Separate ADD and REMOVE heads score candidates, a STOP head and a value head read the context; logits are masked. The policy (≈38k parameters) never sees a pair index, so it applies to any candidate set. We train with PPO [10] on batches of 64 episodes sampled over tasks and budgets $n \in \{3, 5, 8, 13\}$, masking a random 20% of pairs as augmentation; REINFORCE [11] is an ablation. **Deployment.** Label-free: a greedy rollout, or the best of 16 sampled rollouts ranked by J_{rec}; no labels and no J_{sup} calls, and the *final* set is returned. Few-shot (J_{sup} observable, budget B): best-of-k sampling for add-only, and Monte-Carlo look-ahead for add/remove, which evaluates the top removals with sampled replacements and applies the best improving swap.

IV. Experimental protocol

Data. Four annotated ~~multi-excitation fluorescence datasets acquired with a tunable source and a liquid-crystal tunable filter (~~datasets from one system (WheeLED WLS-23-A switchable LED source, Nuance FX tunable-filter camera, emission 420–720 nm in 10 nm steps)~~:~~. ~~*Lichens1*~~*Lichens 1* (~~1040×925 px,~~ 8 excitations 310–430 nm, $M = 192$~~, 4 classes, 16 specimens, public [12]),~~) is columns 466–1390 of the palette in [12]: lower and upper surface of *Ramalina sinensis*, ~~*Lichens2(the same lichens at camera resolution with different processing*~~*Flavopunctelia soredica* and *Ramalina fraxinea*, four pieces each. *Lichens 2* (8 excitations, $M = 210$~~, 5 classes,~~) is a separate, undeposited acquisition of 28 ~~specimens) , and two collagen digestion series,~~ pieces in five annotated classes whose taxa are not recorded [taxa; shared thalli with Lichens 1?]. *Collagen-AA* ~~(acetic acid, $M = 149$) and~~ and *Collagen-P* (~~pepsin~~6 excitations 310–400 nm, ~~$M = 158$) , each with 3 classes~~$M = 149$ and ~~9 specimens~~158 [13]) are nine collagen sponges each (acetic-acid or pepsin extraction) with three crosslinker levels as classes. M counts the pairs left after removing bands within 40 nm (Collagen-P: 30 nm) of λ^{ex} and $2\lambda^{\mathrm{ex}}$. A specimen is a connected annotated region, one physical piece except in the fifth Lichens 2 class (two fragments of one piece); it is the unit of independence. The processed files label the emission axis of a few excitations 15–40 nm off the recorded axis; this changes no feature or selected set, only the wavelength-position features (S1 and the with-wavelength S2 variants) and the axis of Fig. 3, which we plot on recorded wavelengths. **Nested, specimen-grouped evaluation.** Below, the datasets are called Lichens 1, Lichens 2, Collagen-AA and Collagen-P. Outer folds hold out at least one whole specimen per class (~~$k = 4, 2, 3, 3$~~ $k = 4, 2, 3, 3$ folds; three ~~seeded repeats)~~repeats, i.e. re-drawn outer splits; S1 and S3 use repeat 0, S2 all three). A *seed* is the randomness of a search run or of policy training: 3 for the classical wrappers, 1 for per-instance RL, 2 for the non-augmented transferred policies [14], [15]. Every selection method sees only the training specimens; J_{sup} uses inner whole-specimen folds on a per-specimen capped subsample (≈3000 px). Selected sets are scored on the held-out specimens by kNN ($k = 5$, primary, as in ~~our earlier work~~ earlier work on these data), shrinkage LDA, an RBF-SVM and a random forest; we report balanced accuracy (BA) and macro-F1, the number of distinct excitations, mean $|\mathrm{corr}|$, and selection stability (Kuncheva index [16]). **Methods.** Label-free: random (10 draws), top variance, a uniform excitation–emission grid, SPA [17], greedy J_{rec} (column-subset greedy, CSS) and its whitened-PCA variant, and a variance-relevance/correlation-penalty rule mimicking ~~our earlier selector~~ an earlier selector for these data. Supervised filters: Fisher score and mRMR [18]. Wrapper search on J_{sup} with a budget of B uncached ~~evaluations~~J_{sup} evaluations (every scored set counts, partial sets included): random search, sequential forward selection (SFS), SFS followed by best-improvement swaps, beam search

TABLE I

SEARCH AT A MATCHED BUDGET (S1, $B = 5000$, MEAN OVER THE 4 DATASETS AND OF THE REPEAT-0 FOLDS AND SEEDS) AND ABLATIONS (COLLAGEN-AA, $n = 8$, $B = 20000$, ORACLE COST CHARGED). J_{sup} IS THE INNER OBJECTIVE; KNN BA IS ON HELD-OUT SPECIMENS. †BUDGET TRUNCATION (SEE TEXT), NOT A SEARCH RESULT. ‡POLICY TRAINED PER INSTANCE WITH 10^4 EVALUATIONS, THEN LOOK-AHEAD WITH THE REST (10 158 EVALUATIONS IN TOTAL ON AVERAGE, HALF THE BUDGET OF THE OTHER ROWS).

	$n = 5$		$n = 13$		$n = 3$
Method	J_{sup}	kNN BA	J_{sup}	kNN BA	gap to exact
RL add-only	0.864	0.826	0.883	0.841	n/r
RL add/remove	0.861	0.813	0.887	0.844	n/r
SFS+swap	0.871	0.835	0.916	0.853	0.009
Beam+swap	0.872	0.836	0.845†	0.819†	0.008
SA	0.884	0.838	0.920	0.856	0.000
GA	0.885	0.835	0.924	0.860	0.000
Random search	0.854	0.817	0.885	0.839	0.020

Exact optimum at $n = 3$: mean $J_{\text{sup}} = 0.869$ (budget 5000 per method; mean over datasets). n/r: RL not run at $n = 3$.
† Budget exhausted before a complete 13-set in 12/12 tasks (completed with CSS; 5.7 of 13 pairs on average).

Ablation: Collagen-AA, $n = 8$, $B = 20000$, 3 outer folds, 1 seed

	J_{sup}	kNN BA
RL add-only	0.869	0.764
+ oracle prior	0.867	0.759
RL add/remove	0.870	0.762
+ oracle prior	0.866	0.753
terminal reward	0.870	0.765
REINFORCE	0.839	0.736
SA	0.881	0.767
SFS+swap	0.822	0.735
RL add/rem. + MC look-ahead‡	0.866	0.779

(width 5) with swaps, simulated annealing (SA) [19], a steady-state genetic algorithm (GA), and exact enumeration for $n \leq 3$. If the budget ends before any complete n-set has been scored (SFS+swap when B is below one forward step, $\approx M$ evaluations; Beam+swap at $n = 13$), the best partial set is completed with the label-free CSS rule and its J_{sup} is computed outside the budget. **Regimes.** *S1 search:* all wrappers and per-instance RL on each training task, $B = 5000$, best-so-far set. Every step of an RL rollout scores a set, so 5000 evaluations correspond to only 25–91 (add-only) and 14–42 (add/remove) PPO updates of 16 episodes on average; the 250-update runs of S2 instead train one policy on many tasks with no per-instance budget. *S2 transfer:* policies trained with J_{sup} reward on the source datasets—leave-one-dataset-out (LODO) and cross-family (Lichens→Collagen, Collagen→Lichens)—and deployed label-free on every outer fold of the target. *S3 few-shot:* the ~~transferred~~ LODO policy with J_{sup} observable on the target and $B \in \{25, \ldots, 400\}$. Under LODO the source pool of each lichen target contains the other lichen dataset, and that of each collagen target the other collagen dataset (same sponge design, same session), so S3 and the LODO results of S2 are closer to in-domain than the cross-family results. Budgets $n \in \{3, 5, 8, 13\}$ throughout.

V. RESULTS

Per-instance search (S1, Table I). On a single labelled dataset RL is a black-box optimiser, and a weak one: at $B = 5000$ evaluations SA and GA reach the highest inner J_{sup} (0.920/0.924 at $n = 13$ versus 0.883/0.887 for add-only/add-remove RL~~) and recover~~; add-only RL is even below random search, 0.885) and reach the exact optimum at $n = 3$ ~~on every task,~~ up to a mean gap below 0.001 (SA misses it in 5 and GA in 1 of 36 runs, by at most 0.005), while SFS+swap and beam search stop ~~≈0.01 short~~0.008–0.009 short; RL was not run at $n = 3$. RL needs ≈2900 evaluations to come within 0.005 of its own final value (Fig. 4b). Beam+swap at $n = 13$ (J_{sup} 0.845) is budget truncation, not a search result: its first level costs M evaluations and each further level up to $5M$, so 5000 evaluations do not reach depth 13; in all 12 tasks the beam stops early and 5.7 of the 13 pairs, on average, are added by CSS. Outer-test accuracy differences between all other searchers are small (0.84–0.86 at $n = 13$) because the inner objective is optimistic: its gap to the outer BA is 0.04–0.06 (0.026 for the truncated Beam+swap) and grows with search strength (SA/GA 0.064 at $n = 13$, random search 0.046), the winner's curse of selecting on few specimens [15]. Among RL design choices PPO is clearly better than REINFORCE (0.870 vs 0.839 inner J_{sup}), exact pair-enumeration priors do not repay their 11k-evaluation cost, terminal and shaped rewards end at the same value, and a policy trained with half the budget (10^4 evaluations) followed by Monte-Carlo look-ahead ~~gives the best outer accuracy of all methods~~ (about 160 further evaluations) gives the highest outer accuracy on Collagen-AA (0.779 ~~) at half the budget~~vs 0.767 for SA with the full $B = 20000$; 3 folds and one seed, so within noise).

Label-free transfer (S2, Fig. 1, Table II). This is the regime the reversible action was meant for, and the add/remove policy is consistently better than the matched add-only policy: pooled over targets, sources and budgets $n \in \{3, 5, 8, 13\}$ the gain is +0.025 BA (95% CI $[0.022, 0.028]$, 837 wins/315 losses over 1152 paired units, i.e. four variants × 288 target/source/n/repeat/fold units); on cell means (32 target×source×n cells) it is ~~+0.020~~ +0.017 to +0.031 for each of the four policy/deployment variants (Holm ~~$p \leq 7 \times 10^{-3}$~~$p \leq 0.008$, ≥21/32 cells). The gain is largest across sample families (+0.024 to +0.045 per variant over its 144 cross-family units, against +0.013 to +0.017 under LODO) where the constructive policy errs most. For the variant of Table II (wavelength-free, best of 16, ~~and on Lichens1~~ cross-family) it is +0.024 pooled, and +0.017 as the unweighted mean of the 12 tabulated cells ($n = 5$ is not shown); on Lichens 1 it holds in all 12 units at every n (+0.027 to +0.062 for this variant). Removing the wavelength-position features improves every policy (~~+0.014~~ +0.013 to +0.026, $p < 10^{-5}$): absolute or relative wavelengths encode the chemistry of the source and mislead on another sample type. In absolute terms, ~~however, a transferred~~ CSS is strong at $n = 3$ (best label-free value on three of four targets) but not at $n = 13$, where it is below random selection on three of four targets (Lichens 1 0.898 vs 0.948, Collagen-AA 0.655 vs 0.727, Collagen-P 0.721 vs 0.785), and at $n = 8$ on Collagen-AA (0.688 vs 0.699). The transferred add/remove policy alone rarely beats ~~the label-free greedy CSS baseline, which is very strong at small n; it does at~~ CSS significantly: its mean exceeds CSS in 5 of the 12 cells of Table II and significantly in one (Lichens 1, $n = 13$~~on Lichens1 (~~, +0.042, 12/12 units), while it is significantly worse at Lichens 1, $n = 3$ (−0.021). **Task augmentation**

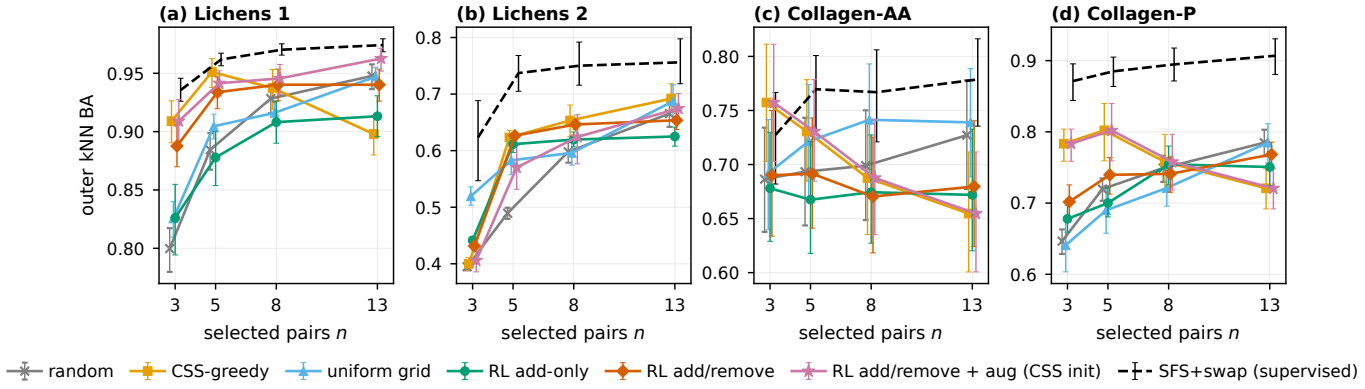


Fig. 1. Label-free transfer to an unseen sample type (S2). Policies are trained with J_{sup} reward on the other family (Collagen→Lichens for (a,b), Lichens→Collagen for (c,d)) and deployed without labels (best of 16 rollouts ranked by J_{rec}). Outer-test kNN balanced accuracy, mean and 95% bootstrap CI over 3 repeats × outer folds (RL: 2 policy seeds, +aug 1). Dashed: supervised SFS+swap on the target~~(upper~~, a supervised reference ~~)~~that uses labels and a search budget on the target; it is not an upper bound.

and CSS refinement (S2). To enlarge the small source pools we retrained every policy on augmented tasks (class subsets, excitation dropout; 42–105 tasks per pool, one seed; 250 ~~instead of 400 updates, one seed~~PPO updates, as for the non-augmented policies, except 400 in the training logs of the cross-family add-only and add/remove ones). The reversible advantage replicates on these independently trained policies: add/remove beats add-only by $+0.019$ cross-family (86/58 units, unit-level Wilcoxon $p = 0.005$; $+0.023$, $p = 10^{-7}$ with best-of-16) and by $+0.027$ under LODO (114/30, $p = 10^{-13}$). Augmentation itself did not help~~(add-only $+0.002$;~~ ~~:~~ against the non-augmented policies it changes add/remove ~~-0.024~~ by -0.014 cross-family ~~, $p = 4 \times 10^{-6}$,~~ ($p = 2 \times 10^{-4}$ greedy, 4×10^{-6} best-of-16) and by $+0.002$/-0.001 under LODO (n.s.~~ under LODO), although shorter training confounds this~~), and add-only by $+0.011$ (greedy)/-0.013 (best-of-16) cross-family; the unequal training length is a confound. Mixing the label-free objective into the training reward ($\Delta J_{\text{sup}} + 0.5\,\Delta J_{\text{rec}}$) recovers $+0.013$ cross-family ($p = 0.001$) but costs 0.007 under LODO. Started from the CSS set ("+aug", Table II) the policy is a conservative refiner, not a reliable one: the collagen-trained policy edits it on the lichen targets, improving it on ~~Lichens-1~~ Lichens 1 (0.937→0.946 at $n = 8$, 11/12 units; 0.898→0.963 at $n = 13$, 12/12, above random at 0.948~~) but~~ ; but -0.010 at $n = 5$, 5/12 units improved) and lowering it on ~~Lichens-2 (-0.02 to -0.05 at $n \geq 5$~~Lichens 2 (-0.029 at $n = 8$, -0.017 at $n = 13$; -0.053 at $n = 5$, not tabulated). The lichen-trained policy and every LODO policy stop at once and return CSS unchanged.

Few-shot search (S3, Fig. 2, Table III). When a few labelled evaluations are allowed on the target, the transferred policy is a strong prior: ~~with only~~ at $B = 25$ ~~evaluations~~ both RL variants match SA (~~pooled differences~~ differences to SA pooled over the four datasets, $n \in \{3, 5, 8, 13\}$ and 48 fold units: $+0.007$ for add/remove with look-ahead and 0.000 ~~, for~~ for add-only best-of-k, both n.s.)~~and~~. At $n = 8$ (Table III) the add/remove policy with look-ahead ~~gives~~ has the best accuracy at ~~every budget on Lichens1, Lichens2 (up to $B = 100$)~~ $B = 25$ and 100 on Lichens 1, Lichens 2 and Collagen-AA ~~, where it also overfits less (lower inner J_{sup}, higher~~ and at $B = 400$ on Collagen-AA (on Lichens 1 at $B = 400$ it is within 0.001 of add-only). It is not generally better at the inner objective: averaged over the four datasets at $n = 8$, SA has the higher inner J_{sup} than add/remove at $B = 100$ (0.876 vs 0.849) for similar outer BA (0.836 vs 0.833), while at $B = 25$ SA has 0.831 vs 0.824 inner but 0.794 vs 0.827 outer~~BA)~~. Classic search catches up as the budget grows~~;~~ ~~:~~ at $B = 400$ SA is ahead ~~by~~ of the add/remove and add-only policies by 0.024 and 0.017 ~~-0.024 (Holm $p = 0.008/0.08$~~(same pooling as above; Holm $p = 0.08/0.008$; 0.009 and 0.016 at $n = 8$ alone), driven by Collagen-P, where GA ~~/SA reach 0.92 and the policy plateaus~~ and SA reach 0.920 and 0.908 and the policies plateau at 0.82–0.85. Look-ahead versus best-of-k add-only differs by ~~less than~~ at most 0.008 pooled over datasets, n and budgets (n.s.)~~.~~; at $n = 8$ alone the difference is $+0.028$, $+0.030$ and $+0.007$ at $B = 25$, 100, 400, concentrated on Lichens 2 ($+0.078$ at $B = 25$). SFS+swap is a supervised reference, not an upper bound: for B below one forward step ($\approx M$ evaluations) it fixes one pair by SFS and fills the other 7 with CSS ($B = 25$, 100), 6 at $B = 200$ and 5 at $B = 400$, which is why its accuracy does not increase with B (Collagen-AA 0.716→0.688).

Is the reversible policy harder to train? No. Despite the longer episodes and the delayed credit of a swap, add/remove policies reach within 0.01 of their final training return ~~after 112–123~~ (mean of the last 20 updates; first update at which the 10-update moving average gets there) after 108 PPO updates versus ~~153–166~~ 143 for add-only without wavelength features, and 138 versus 153 with them (Fig. 4a), with a comparable final value (~~0.834 vs 0.829 without, 0.837~~ 0.833 vs 0.824 without, 0.846 vs 0.839 with wavelength features). ~~Paired credit for the remove–add cycle, the tabu maskand~~ We did not ablate the tabu mask, the STOP action ~~keep the refinement stable~~or the revisit penalty.

TABLE II

LABEL-FREE TRANSFER (S2), OUTER KNN BA (MEAN OVER 3 REPEATS × FOLDS; $n = 5$ OMITTED). RL COLUMNS: WAVELENGTH-FREE FEATURES, BEST OF 16 ROLLOUTS BY J_{rec}, CROSS-FAMILY SOURCE (COLLAGEN→LICHENS FOR THE LICHEN ROWS, LICHENS→COLLAGEN FOR THE COLLAGEN ROWS). "+AUG": TRAINED WITH CLASS-SUBSET AND EXCITATION-DROPOUT TASK AUGMENTATION, REFINING THE CSS SET (IDENTICAL TO CSS WHERE THE POLICY STOPS AT ONCE). p_{add}: HOLM-CORRECTED WILCOXON p OF ~~ADD/REMOVE~~ THE *RL add/rem* COLUMN VS ~~ADD-ONLY (SIGN = DIRECTION)~~THE RL ADD COLUMN; p_{CSS}: ~~ADD/REMOVE~~ *RL add/rem* VS CSS; SIGN + MEANS RL ADD/REM IS HIGHER. $^*p < 0.05$. BOLD: BEST LABEL-FREE VALUE (TIES AT THREE DECIMALS ARE ALL BOLD).

Target	n	Random	CSS	Best other	RL add	RL add/rem	RL +aug	p_{add}	p_{CSS}
Lichens 1	3	0.800	**0.909**	0.826 (grid)	0.826	0.888	**0.909**	+.012*	−.012*
	8	0.928	0.937	0.916 (grid)	0.908	0.940	**0.946**	+.012*	+1.00
	13	0.948	0.898	0.947 (grid)	0.913	0.940	**0.963**	+.012*	+.012*
Lichens 2	3	0.395	0.400	**0.519** (grid)	0.442	0.431	0.406	−.625	+.438
	8	0.598	**0.653**	0.596 (grid)	0.620	0.646	0.624	+.438	−1.00
	13	0.665	**0.692**	0.687 (grid)	0.625	0.654	0.675	+.438	−1.00
Collagen-AA	3	0.686	**0.757**	0.746 (var)	0.678	0.690	**0.757**	+1.00	−.074
	8	0.699	0.688	**0.743** (var)	0.674	0.670	0.688	−1.00	−1.00
	13	0.727	0.655	**0.743** (var)	0.672	0.680	0.655	+.141	+.188
Collagen-P	3	0.647	**0.783**	0.641 (grid)	0.678	0.702	**0.783**	+.602	−.141
	8	0.749	**0.758**	0.721 (grid)	0.754	0.741	**0.758**	−1.00	−1.00
	13	**0.785**	0.721	**0.785** (grid)	0.751	0.768	0.721	+1.00	+.293

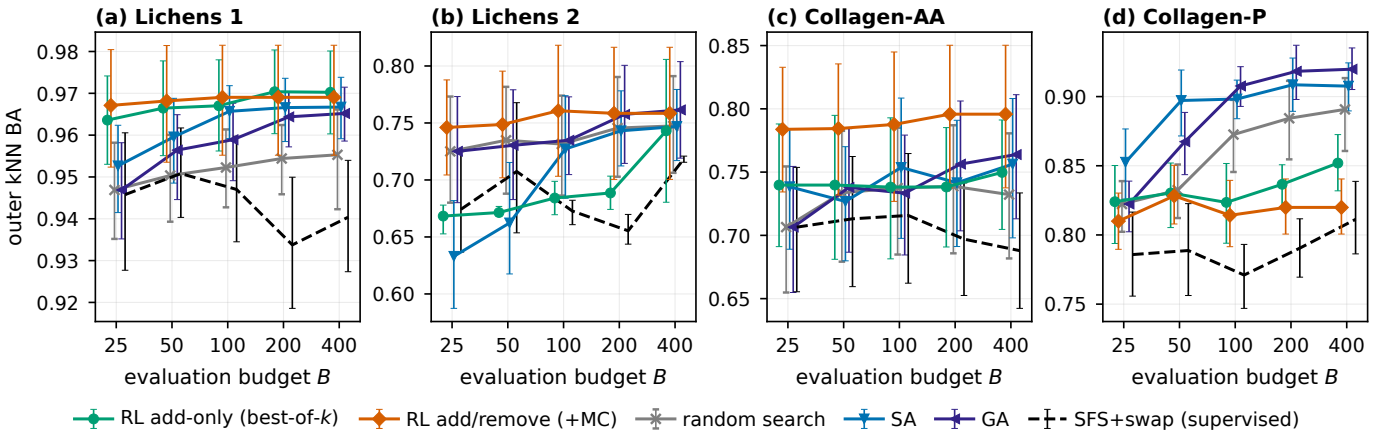


Fig. 2. Few-shot search on the target (S3, $n = 8$): outer kNN BA vs. the number of labelled objective evaluations B. RL policies are the leave-one-dataset-out policies transferred to the target; mean and 95% CI over repeat-0 outer folds and seeds. SFS+swap is a supervised reference, not an upper bound (see text).

Which pairs, and at what hardware cost? Fig. 3 shows that label-free CSS returns almost the same few pairs in every fold, whereas the reversible policy and the supervised searchers spread their choices over more excitations, mostly 365–430 nm, and over the short- and long-wavelength ends of the emission range. ~~Adding a cost μ~~ In a separate per-instance experiment on Lichens 1 ($n = 8$, $B = 5000$, 4 outer folds of repeat 0, one seed), adding a cost $\mu \in \{0, 0.005, 0.01, 0.02, 0.05\}$ per distinct excitation to ~~the reward reduces the~~ J_{sup} reduces the mean number of light sources from ~~5.3 to 1.0 for the~~ 5.25 ($\mu = 0$) to 1.00 ($\mu = 0.05$) for the add/remove RL policy (~~4.8 to 1.5~~ 4.75 to 1.50 for SA), while kNN BA ~~on Lichens1 stays at 0.93–0.97~~is 0.970 at $\mu = 0$ and 0.931 at $\mu = 0.05$ for RL (SA: 0.970 and 0.915); four folds give no reliable intervals.

VI. DISCUSSION AND CONCLUSION

The study answers the question that motivated it with a qualified yes. The ability to remove and swap pairs makes a transferred selection policy consistently better than an otherwise identical add-only policy, by 2–3 BA points on average and up to 6 points, and the advantage is largest when the policy meets a sample type unlike its training data. This supports the intuition that undo actions matter most where a constructive policy is miscalibrated [6]. The reversible policy was not harder to train: ~~shared swap credit, tabu masking and a stop action made refinement converge~~ it reached its plateau in fewer updates than ~~construction~~the add-only policy without wavelength features and in about as many with them.

Three negative results are equally important for practice. First, when the selection objective can be evaluated cheaply on the target, RL is not the right tool: SA and GA find better subsets at the same budget, and they reach the certified

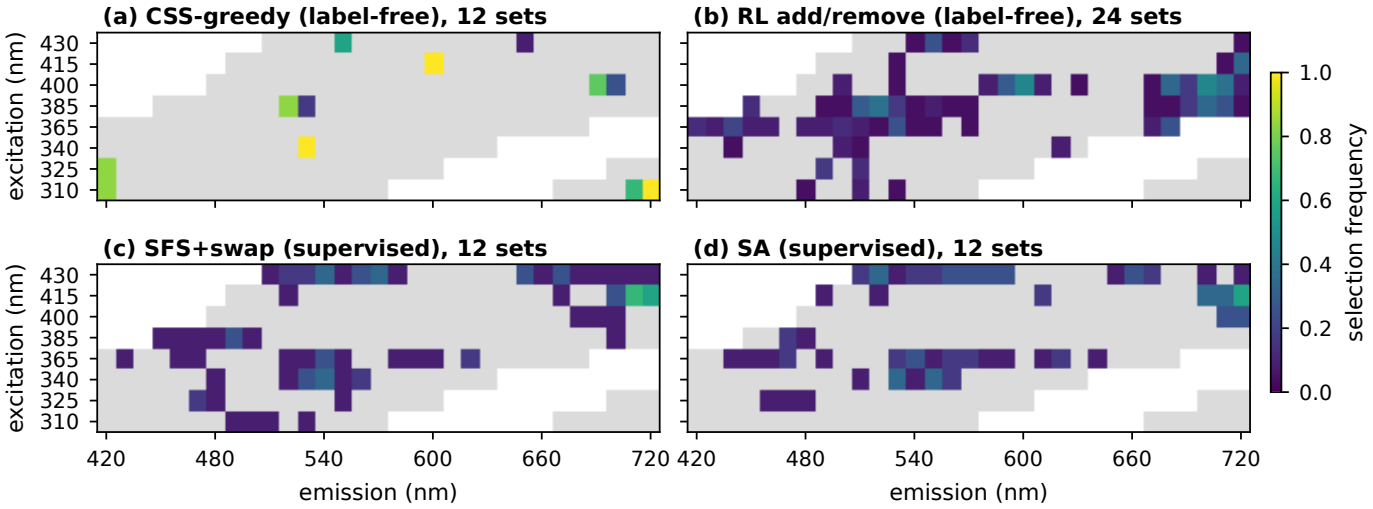


Fig. 3. Selection frequency of excitation–emission pairs on ~~Lichens1~~ Lichens 1 ($n = 8$) over all outer folds and repeats; grey cells are admissible pairs never selected. Sets pooled: (a), (c) 12 = 4 folds × 3 repeats; (b) 24 = 2 policy seeds × 12; (d) 12 = 4 folds × 3 search seeds (S1, repeat 0).

TABLE III
FEW-SHOT SEARCH (S3), $n = 8$: OUTER KNN BA (MEAN OVER REPEAT-0 FOLDS AND SEEDS) FOR BUDGETS B OF LABELLED EVALUATIONS ON THE TARGET. SFS+SWAP: SUPERVISED REFERENCE; AT THESE B MOST OF ITS PAIRS ARE CSS-FILLED (SEE TEXT).

Dataset	B	RL add (BoK)	RL add/rem (MC)	Random	SA	GA	SFS+swap
Lichens 1	25	0.964	**0.967**	0.947	0.953	0.947	0.946
	100	0.967	**0.969**	0.952	0.966	0.959	0.947
	400	**0.970**	0.969	0.955	0.967	0.965	0.940
Lichens 2	25	0.668	**0.746**	0.725	0.633	0.725	0.674
	100	0.684	**0.761**	0.731	0.727	0.735	0.673
	400	0.743	0.758	0.747	0.747	**0.761**	0.719
Collagen-AA	25	0.740	**0.784**	0.707	0.738	0.707	0.706
	100	0.738	**0.788**	0.737	0.754	0.733	0.716
	400	0.750	**0.796**	0.732	0.756	0.764	0.688
Collagen-P	25	0.824	0.810	0.822	**0.853**	0.822	0.786
	100	0.824	0.814	0.873	0.898	**0.908**	0.771
	400	0.852	0.820	0.891	0.908	**0.920**	0.811

optimum at $n = 3$. Second, the inner objective overfits few specimens, so strong search buys little on held-out specimens; reporting the inner–outer gap should be standard. Third, a label-free greedy column-subset rule is a strong baseline at small budgets that most published band selectors do not report ~~;~~ (though it is below random selection at $n = 13$ on three of four targets), so a learned policy should be judged against it and against random selection. Using the policy to refine that greedy set is ~~safe, because it mostly stops, but it paid off on only one of four~~ not reliably safe: it helped on Lichens 1, lowered accuracy on Lichens 2, and stopped immediately on the collagen targets. Wavelength-position features hurt transfer, which suggests that what transfers between sample types is a statistical selection rule, not a spectral prior.

Limitations. We had four datasets from two sample families and one instrument, so the source pools for transfer are small (two or three datasets); task augmentation by class subsets and excitation dropout only partly substitutes for more acquisitions. Specimen counts are low (three per class for collagen), which makes every outer estimate noisy. Simulation-based training, for example in a physics-based simulator, and sequential acquisition where the policy controls the instrument are natural next steps.

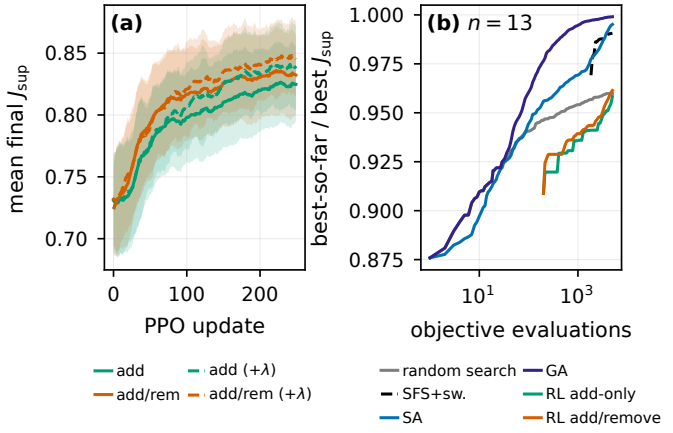


Fig. 4. (a) PPO training curves (mean final J_{sup} of training episodes, mean ~~and range~~ ± s.d. over the six source pools, two seeds each for the wavelength-free policies): add/remove reaches its plateau earlier than add-only; dashed (+λ): with wavelength features (250 updates for all). (b) S1 anytime curves at $n = 13$: best-so-far J_{sup}, normalised per task by the best value found by any method.

Conclusion. Reversible add/remove/stop actions are a simple, robust improvement for RL-based spectral pair selection on new data, but classic search should remain the default for a fixed labelled dataset, and a label-free greedy ~~selection is the baseline to beat~~baseline should be reported alongside random selection.

REFERENCES

{url@samestyle
[1] J. R. Lakowicz, *Principles of Fluorescence Spectroscopy*, 3rd ed.1em plus 0.5em minus 0.4emSpringer, 2006.

[2] I. Bello, H. Pham, Q. V. Le, M. Norouzi, and S. Bengio, "Neural combinatorial optimization with reinforcement learning," *arXiv:1611.09940*, 2016.

[3] E. Khalil, H. Dai, Y. Zhang, B. Dilkina, and L. Song, "Learning combinatorial optimization algorithms over graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[4] W. Kool, H. van Hoof, and M. Welling, "Attention, learn to solve routing problems!" in *Int. Conf. Learning Representations (ICLR)*, 2019.

[5] L. Mou, S. Saha, Y. Hua, F. Bovolo, L. Bruzzone, and X. X. Zhu, "Deep reinforcement learning for band selection in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[6] T. D. Barrett, W. R. Clements, J. N. Foerster, and A. I. Lvovsky, "Exploratory combinatorial optimization with reinforcement learning," in *AAAI Conf. Artificial Intelligence*, 2020.

[7] X. Chen and Y. Tian, "Learning to perform local rewriting for combinatorial optimization," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[8] C. Boutsidis, M. W. Mahoney, and P. Drineas, "An improved approximation algorithm for the column subset selection problem," in *ACM-SIAM Symp. Discrete Algorithms (SODA)*, 2009.

[9] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Poczos, R. Salakhutdinov, and A. Smola, "Deep sets," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[10] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv:1707.06347*, 2017.

[11] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Machine Learning*, vol. 8, pp. 229–256, 1992.

[12] N. Chilingaryan and N. Sarvazyan, "250707_4dhsi_lichens_palette," Zenodo, doi:10.5281/zenodo.18640119, 2026.

[13] K. Ghahramanyan, "250929_collagen," Zenodo, doi:10.5281/zenodo.20677504, 2026.

[14] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[15] S. Varma and R. Simon, "Bias in error estimation when using cross-validation for model selection," *BMC Bioinformatics*, vol. 7, p. 91, 2006.

[16] L. I. Kuncheva, "A stability index for feature selection," in *IASTED Int. Multi-Conf. Artificial Intelligence and Applications*, 2007, pp. 390–395.

[17] M. C. U. Araújo, T. C. B. Saldanha, R. K. H. Galvão, T. Yoneyama, H. C. Chame, and V. Visani, "The successive projections algorithm for variable selection in spectroscopic multicomponent analysis," *Chemometrics and Intelligent Laboratory Systems*, vol. 57, no. 2, pp. 65–73, 2001.

[18] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, 2005.

[19] S. Kirkpatrick, C. D. Gelatt, and M. P. Vecchi, "Optimization by simulated annealing," *Science*, vol. 220, no. 4598, pp. 671–680, 1983.